AI-Powered Digital Marketing Orchestration: An Integrated, Closed-Loop Framework for Cold-Start Digital Marketing

Segmentation · Campaign Automation · Attribution-Aware Analytics · AI Content Refinement

Problem in Brief

- Digital marketing is executed as **disconnected actions, not as a learning process** — one tool segments, another automates, a third reports, a fourth writes, and none exchange information.
- The difficulty is sharpest for a **newly launched product**, where almost no behavioural history exists and data-hungry models are least effective.
- **Segmentation collapses under low data** — clustering cannot express "there is not enough evidence about this person."
- **Policies are asserted, not compared**, and attribution models are only ever compared against each other.
- **The system records no basis for its own improvement** — a deterministic policy assigns probability zero to actions it never takes.

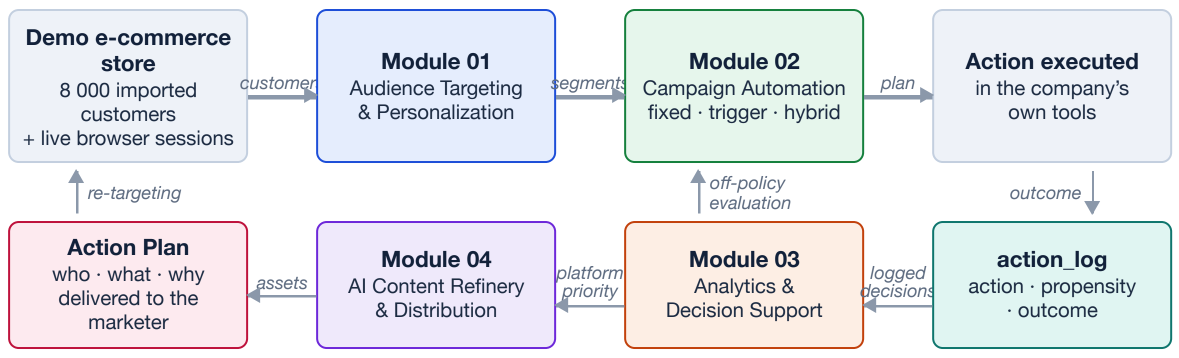
Aim

To design, implement and experimentally evaluate an integrated, closed-loop AI marketing framework unifying segmentation, automation, attribution-aware analytics and content refinement — effective under cold start, and **whose every reported claim is reproducible from the system itself.**

Objectives

- Build a hybrid segmentation engine and **ablate it against its own components.**
- Compare three automation policies over paired trials, **cost beside outcome.**
- Compare four attribution models against ground truth where one exists, and quantify disagreement where it does not.
- State what prediction quality **transfers** to another audience.
- Test whether simpler models are competitive, and whether the engagement corpus holds signal.
- Make recommendations **evaluable** by logging every decision with its propensity.
- Integrate the four modules, and make overstating structurally difficult.

Overall System Design



The loop closes twice — once through analytics feedback into segmentation, automation and content, and once through the propensity-logged decision record, which is what makes the policy itself evaluable. The system is an **advisor, not a sender**: it hands over an Action Plan a company executes in its own tools.

Module 01 — Audience Targeting and Personalization

215110N · Sarah M. M. F.

This module segments a cold-start audience by combining a rule-based segmenter with k-means and Ward-linkage hierarchical clustering over one feature matrix. An agreement vote across the three yields both a segment and a **calibrated confidence**, and preserves a *New Cold User* segment that clustering alone cannot represent.

Literature Review

- Segmentation moved from demographic rules to behavioural clustering; k-means and hierarchical methods are the standard treatment (Han et al.; Aggarwal).
- Reviews of e-commerce personalisation find cluster-quality metrics dominate evaluation, while downstream marketing outcomes are rarely reported (Alves Gomes & Meisen).
- Hybrid rule-plus-clustering approaches are proposed to keep interpretability while gaining adaptability.
- Cold start is characterised for recommending items to users (Schein et al.), but not for segmenting users who have none.

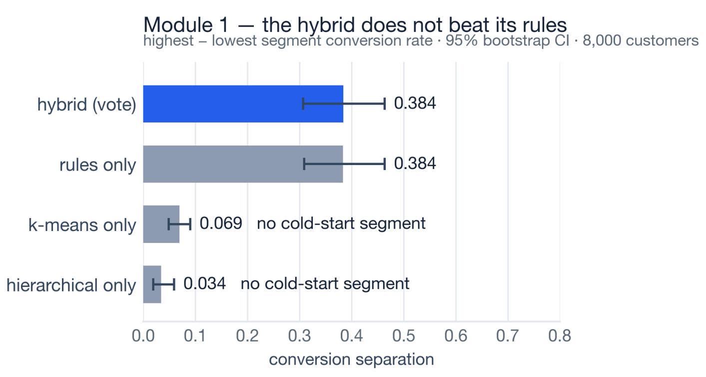
Research Gaps

- Hybrids are rarely measured **against their own components** — the ablation is not run.
- Arbitration is unspecified: when the rule and the cluster disagree, which one wins?
- Cold start is treated as a data problem, not as something a method must be able to *represent.*
- Evaluation stops at cluster quality and never reaches a marketing outcome.

Methodology

Imported audience + live sessions
▼
Behavioural feature matrix
▼
Rule segmenter | k-means | Ward hierarchical
▼
Agreement vote
cold-start rule tested first
▼
Calibrated confidence · Random Forest
▼
Five segments → **user_segments**

Evaluation & Results



Separation **0.3838** (hybrid) against **0.3837** (rules alone); k-means 0.069, hierarchical 0.034. **A negative result, reported as the headline** — the clustering contributes nothing to separation, and the hybrid earns its place on calibrated confidence and explicit cold start instead.

Future work

- Re-fit segment thresholds on a live audience rather than an imported one.
- Extend the vote beyond three methods, with weighted agreement.
- Test the cold-start rule on a second product category.

Module 02 — Marketing Automation and Campaign Management

215129F · Zanar M. H. M. R. A.

Three campaign policies — fixed workflow, behavioural trigger and hybrid — run behind a **single policy engine that is the only component differing between arms.** Thirty paired seeds put the same 8 000 customers through all three, so any measured difference is attributable to the policy alone.

Literature Review

- Fixed-workflow campaign management delivers identical sequences regardless of behaviour, producing low engagement (Chaffey & Ellis-Chadwick).
- Behaviour-triggered automation improves relevance at the cost of growing rule complexity (Davenport et al.).
- Blended approaches modulate a stable base workflow by predicted engagement probability (Kumar & Reinartz).
- Reinforcement learning suits sequential campaign decisions but needs extensive interaction history, restricting it to warm start (Sutton & Barto).

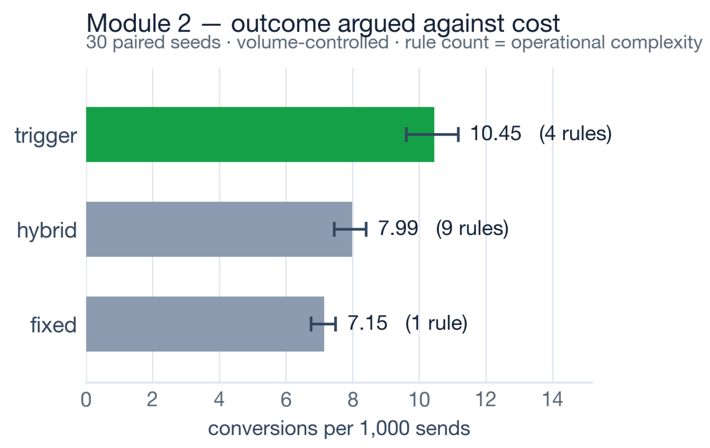
Research Gaps

- Controlled comparison is rare — policies are seldom run on the same users, so reported differences **confound policy with population.**
- Operational cost is almost never measured alongside effectiveness, although it decides the engineering choice.
- Per-customer conversion rate rewards sending *more*, not sending *better.*

Methodology

Segments + confidence (M1)
▼
Shared profile builder
▼
Policy engine — the only arm that differs
▼
fixed 1 rule | trigger 4 rules | hybrid 9 rules
▼
Shared response simulator · 30 paired seeds
▼
Evaluator — conversions / 1 000 sends

Evaluation & Results



Trigger **10.45** against fixed 7.15 — paired difference **+3.299** [3.146, 3.447], Wilcoxon $p = 1.9 \times 10^{-9}$, Cliff's δ = 1.0, bought with three extra decision rules. **The live build ranks the policies differently, and that instability is the result.**

Future work

- A contextual-bandit layer choosing the policy per segment online.
- Response fitted on observed opens rather than simulated propensities.
- Cost extended beyond rule count to send cost and staff time.

Module 03 — Marketing Analytics and Decision Support

215015D · Arqam Z. H.

Funnel analytics, four attribution models, conversion and drop-off prediction, uplift modelling and off-policy evaluation over an append-only decision log. The module turns description into a recommendation — and then makes that recommendation **evaluable from the system's own logs.**

Literature Review

- Funnel analysis locates loss in the journey but is descriptive — it identifies the leak without prescribing the repair (Court et al.).
- First- and last-touch attribution are causally unsound, assigning all credit on position alone (Dalessandro et al.).
- Data-driven multi-touch attribution learns weights from sequences; Markov and Shapley formulations need journey volume (Shao & Li).
- Uplift modelling separates predicting *response* from predicting the *incremental effect* of an intervention (Radcliffe & Surry).

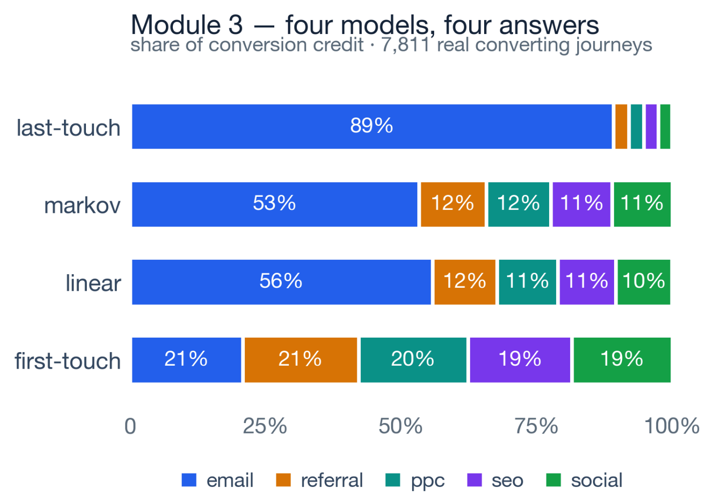
Research Gaps

- Attribution models are compared with one another, **never against known ground truth** — observational data has none.
- Analytics prescribes by ranking predicted response — which is not the question a campaign asks.
- A deterministic policy assigns probability zero to every action it never takes, so its own logs cannot evaluate an alternative.

Methodology

Campaign events (M2) + segments (M1)
▼
Journey-level feature builder
▼
Funnel & drop-off | Attribution ×4 | Uplift S / T
▼
Recommendation generator
+ 10% exploration on purpose
▼
Off-policy evaluation · IPS · SNIPS · DR
▼
action_log → Modules 02 and 04

Evaluation & Results



The four models disagree over **68%** of all attributed credit across 7 811 journeys — last-touch credits email 89%, first-touch spreads it evenly. **The disagreement is the finding, not a defect in one model.**

Future work

- Run the log long enough to close the loop with real decisions — the mechanism is built.
- Doubly-robust estimation once a reward model is trained on the log.
- Live-audience re-fitting so thresholds, not only rankings, transfer.

Module 04 — AI Content Refinery and Multi-Platform Distribution

215001G · Aadhil M. H. M.

The client's own site becomes platform-specific assets — classified for goal and tone, scored on meaning, platform fit and engagement, and checked against **what the site can actually do.**

Literature Review

- Large pre-trained language models made fluent, controllable generation practical (Vaswani et al.; Brown et al.); sentence embeddings check that meaning survives.
- Platforms differ in length, hashtag norms and formality, so one message republished everywhere underperforms.
- Engagement prediction is the weakest link — reported successes are frequently inflated by leakage.

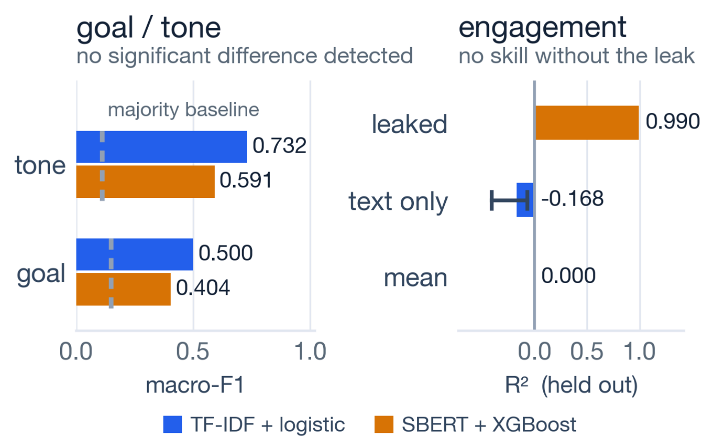
Research Gaps

- Generated content is judged on linguistic quality **in isolation**, disconnected from campaign analytics.
- No check that a recommended action is one the client's site can perform.
- Engagement models are rarely tested for whether the corpus carries any text signal.

Methodology

Client site + brief + Module 03 platform priority
crawler → knowledge base + site capability detection
▼
Campaign goal + brand tone from the knowledge-base text
TF-IDF + logistic vs SBERT + XGBoost → platform register
from the 689-row weak-labelled RafaM97 corpus
▼
BART summary → Phi-3-mini candidates, four platforms
▼
semantic 0.40 | platform fit 0.40 | engagement 0.20
▼
best < 0.75 → regenerate, **goal and tone held fixed**
optimisation re-prompt + paired significance test

Evaluation & Results



TF-IDF + logistic reached higher macro-F1 than SBERT + XGBoost on goal (**0.500** vs 0.404, against a 0.148 majority baseline) and tone (**0.732** vs 0.591) — accuracy **0.772** and **0.818**. Exact McNemar found **no significant difference** in errors (p = 1.000 and 0.727), which is not proof they are equal; TF-IDF is deployed for both, for cost and interpretability. Removing the leaked outcome columns drops engagement prediction from R^2 0.990 to **−0.168**, with **0 of 8 text features surviving Holm–Bonferroni**; its weight was cut 0.45 → 0.20. Capability detection matches hand labels **21 / 21** on a development set.

Preliminary: the held-out sets are small, so these comparisons are underpowered; larger human-reviewed platform-labelled data is needed.

Future work

- A corpus carrying genuine engagement signal; active learning to grow the goal/tone corpora.
- Held-out capability and goal/tone samples rather than development sets.
- A learned platform-suitability classifier replacing the rule checklist.

Conclusion

This research delivers an integrated, closed-loop AI marketing framework in which segmentation, automation, analytics and content generation operate over **one audience, with feedback**, under cold-start conditions. Eleven experiments are reproducible with a single command, and every table and figure is regenerated from the results they write.

Four findings stand on their own: a hybrid must be measured against its own components; **accuracy is not what a leak corrupts** (0.0013 of accuracy, 150× of certainty, every test green); attribution models disagree enough to reverse a spending decision; and a recommender ranking by predicted response answers a different question — so the system logs propensities and explores on purpose. **Stability is not correctness.**

Seven defects were found in the team's own research code — a leaked confidence model, a cold-start segment deleted by its own vote, an engagement regressor trained on its own target. **Every one made a result look better than it was, and none produced an error message.** That is why provenance is derived at write time, propensity is a mandatory column, and there is one regression test per defect.

Not supported by the evidence: that the hybrid separates conversion better than rules alone; that any one automation policy is best; that engagement is predictable from caption text on this corpus. Event ordering is reconstructed, and campaign response is modelled in both simulator and live build.

Final Integration of Modules

PRESENTATION — Next.js 16 · React · TypeScript · Recharts

Marketer dashboard 7 pages | **Auth** login · onboarding | **Demo store** site under study | **Visitor** first-party · DNT

APPLICATION — FastAPI · Uvicorn · Pydantic *↑ HTTPS / JSON · tracking beacon*

Routers segments · campaigns · analytics | **Tracking** /collect · mos.js | **Ownership** tenant isolation | **Decisions** catalogue + propensity | **Provenance** source derived

RESEARCH MODULES — Python · scikit-learn · XGBoost · Phi-3 · BART *↑ invoke*

Module 01 rules + k-means + hierarchical → vote | **Module 02** policy engine + shared simulator | **Module 03** funnel · attribution · uplift · OPE | **Module 04** crawl · goal/tone · generate · score

DATA — PostgreSQL 16 *↑ read / write*

visitors source derived | **user_segments** + confidence | **interactions** source derived | **action_log** propensity · reward | **content_assets** scored assets

11/11	**8 000**	**177**	**7**
experiments, one command	customers, four modules	tests, one per defect	defects found in our own code

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